

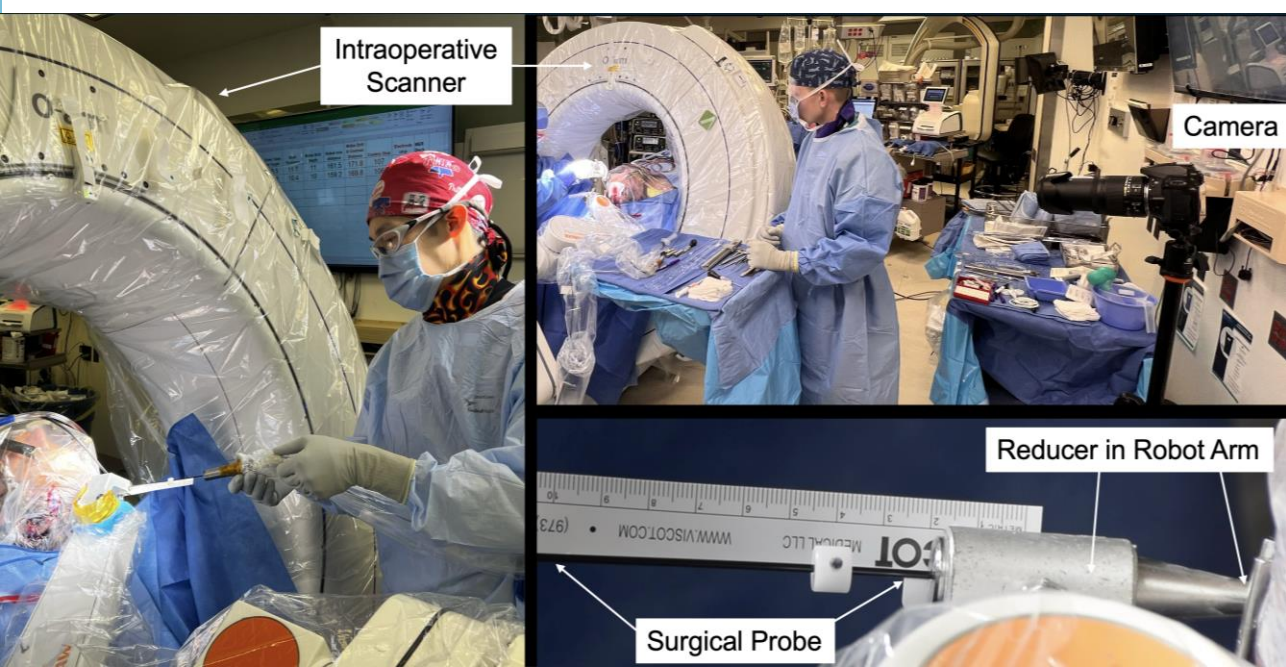
Re-evaluating Sources of Electrode Deviation in Deep Brain Stimulation: Surgical Technique vs. Patient-Specific Brain Mechanics

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Objective

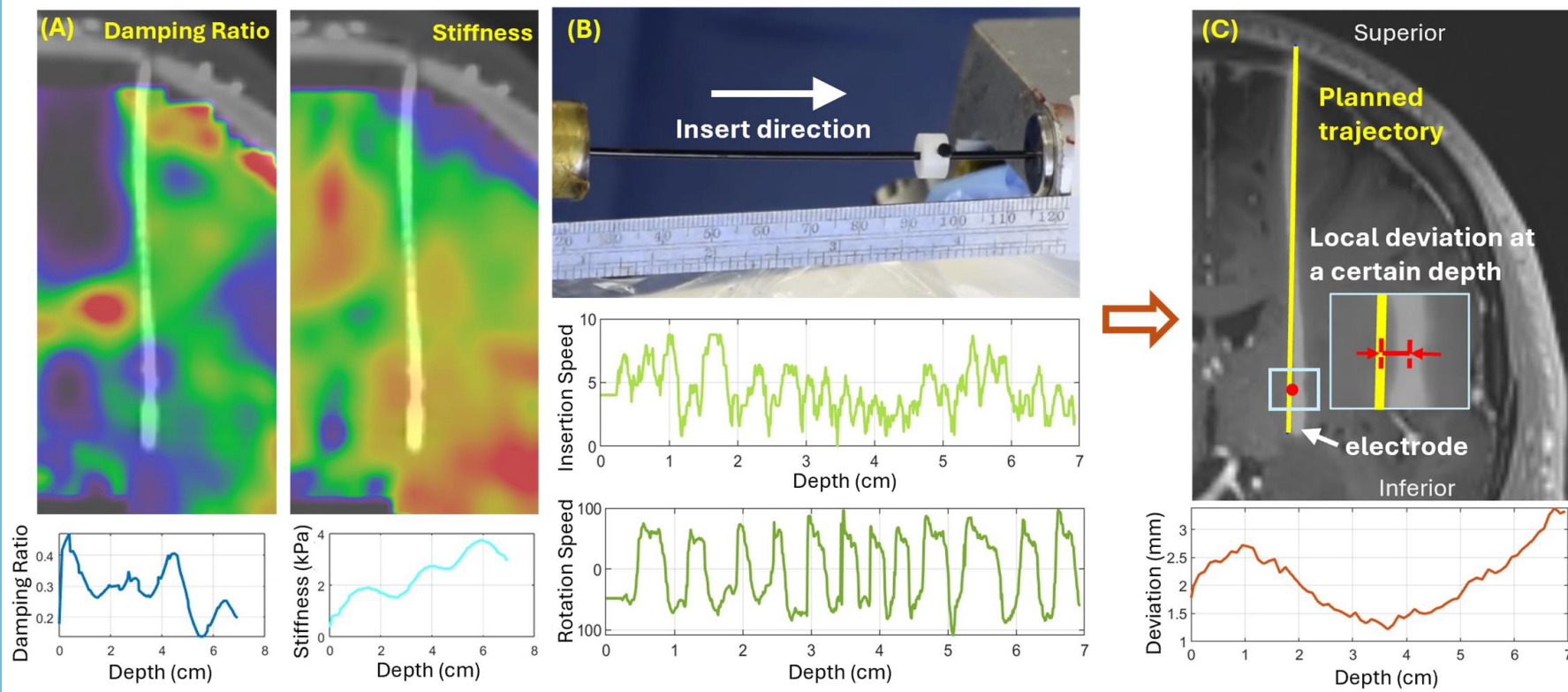
Accurate electrode placement is crucial for optimizing clinical outcomes in **deep brain stimulation (DBS)** for movement disorders. Studies to date have focused on improving surgical techniques to minimize electrode deviation. However, the role of **brain mechanics** is emerging as a key factor in electrode deviation and remains understudied. We sought to quantify the source of deviation from biomechanical sources. **We hypothesize that persistent deviation is explained by patient-specific brain tissue biomechanics.**

Method



Electrodes were implanted using a robotic, image-guided, image-verified technique, consisting of inserting a radiofrequency (RF) probe into the brain to create a channel for the electrode.

Patients with Parkinson's disease undergoing DBS **preoperatively** underwent an MRI and high-resolution **magnetic resonance elastography (MRE)** scan. RF probe **insertion and rotation** were captured by **intraoperative** video, tracking a white depth stop with a black set screw relative to an adjacent ruler. (B) Final electrode location was evaluated on postoperative CT scans, and **local deviations** along the entire intracranial path were measured. (C) **Brain biomechanical properties (stiffness and damping ratio)** along the trajectory extracted from the **coregistered MRE data**, which is aligned with entire electrode deviation. (A)



Results

Recruitment:

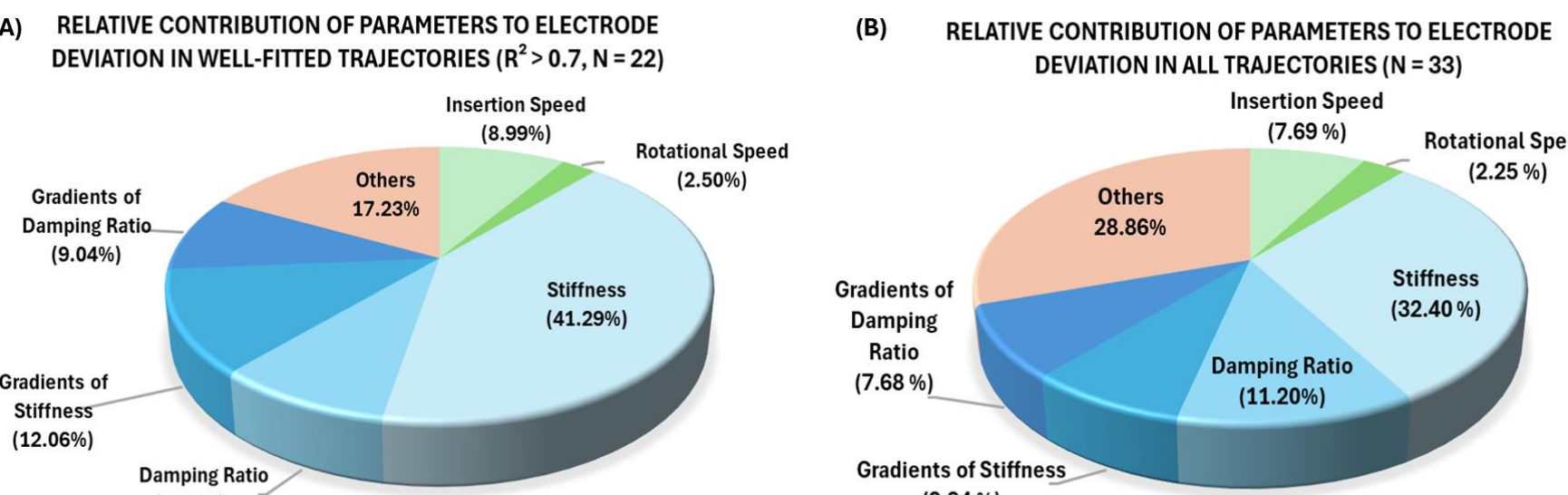
17 patients were enrolled, contributing 41 electrode trajectories, which included revision trajectories to optimize electrode positioning in the same patient. After screening, **33 trajectories** (target deviation of 1.71 ± 1.18 mm; pneumocephalus 5.22 ± 5.42 cc) from 14 patients (a mean age of 62.3 years and an average disease duration of 7.9 years.) were included in the analysis.

Regression analysis:

33 trajectories were analyzed using a consistent multiple linear regression model framework. All trajectory models yielded significant results (model $p < 0.05$) with an **average R^2 of 0.71**.

Factors Contributing to Electrode Deviation

Scaled Partial R^2 is calculated for each predictor in each of the trajectory models. **MRE metrics** explained on average **71.27%** of the local deviation variance across the 22 well-fitted trajectories ($R^2 > 0.7$)



Feature	p-value
Stiffness Features	
Mean stiffness	0.12
Standard deviation	<0.001
Minimum	<0.001
Maximum	0.45
Range	<0.001
Max Gradient	0.03
Area under curve (AUC)	0.10
Damping Ratio Features	
Mean damping	0.27
Standard deviation	0.95
Minimum	<0.001
Maximum	0.94
Range	0.58
Max Gradient	0.45
Area under curve (AUC)	0.35

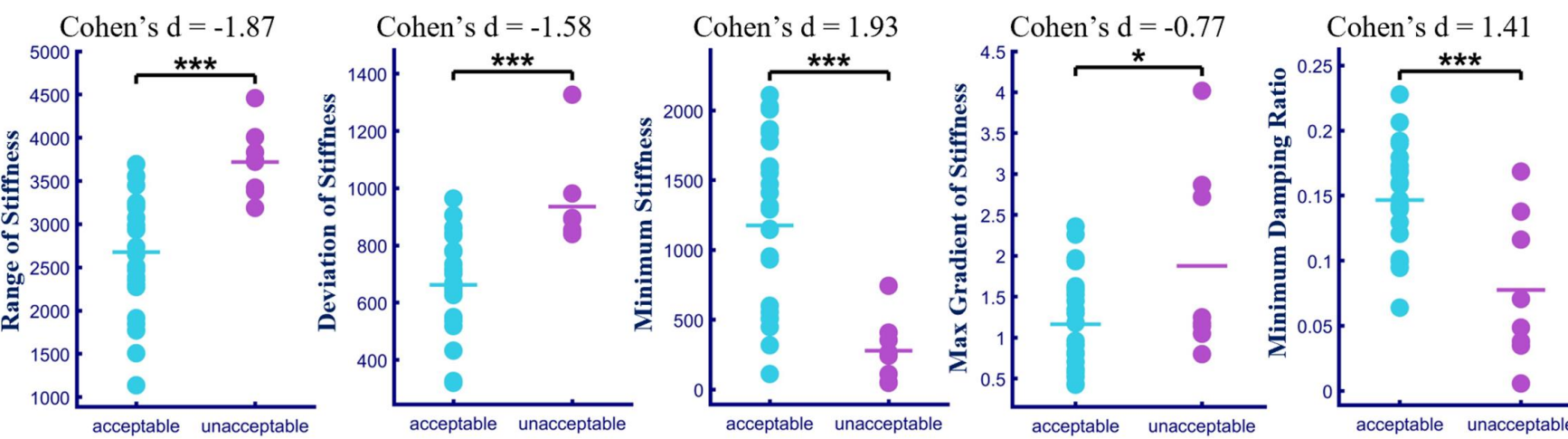
Brain Tissue Properties in Suboptimal Electrode Placements

Independent samples t-tests were conducted to compare features derived from MRE-derived tissue properties between acceptable (< 2 mm) and unacceptable (≤ 2 mm) target deviations.

Feature	p-value
Clinical Variables	
Age at time of surgery	0.53
Years since diagnosis	0.04
Preoperative UPDRS	0.80
Pneumocephalus	0.60

Results

Cohen's d is calculated for parameters with a significant difference between groups of acceptable and unacceptable target deviations. Trajectories with **acceptable target deviations** exhibited a **smaller stiffness range, lower variation (standard deviation) of stiffness, and higher minimum values of both stiffness and damping ratio**. The absolute value of Cohen's d of these four features between the two groups were all **larger than 1.4**. Maximum gradient of stiffness was significantly lower in the acceptable group, but with a relatively smaller effect size.



Conclusion

Despite advances in surgical technique, the underlying factors contributing to electrode deviation remain incompletely understood. Here, we have identified that while procedural contributors are relevant, **patient-specific brain biomechanical properties as measured noninvasively via MRE are the dominant contributors to continued target deviation**. These findings highlight the need for personalized, patient-specific biomechanical informed approaches to improve targeting precision and surgical outcomes.

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